

SMART POWER DISTRIBUTION & CONSUMPTION ANALYTICS SYSTEM

Tutorial:- 3

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Smart Power Distribution & Consumption Analytics System

1. Project Overview

The Smart Power Distribution & Consumption Analytics System is a database driven application designed to model, manage, and analyze electricity distribution and consumption at a city level.

The primary objective of the system is to provide a structured, reliable, and transaction safe relational database solution for electricity distribution companies to track power flow from grids to consumers, record electricity usage through smart meters, generate monthly bills automatically, and perform analytical queries for operations.

Unlike real time power grid simulators or heavy systems, this project focuses on data integrity, consistency, and analytical capability.

The system models a hierarchical distribution structure where electricity flows from **Power Grids** → **Distribution Areas** → **Consumers**, with each consumer equipped with a smart meter. Monthly meter readings are captured and used to automatically generate electricity bills based on a predefined cost. The system also supports area wise consumption analysis and loss detection by comparing electricity supplied versus electricity consumed etc.

Problem Statement

Electricity distribution companies face several operational and analytical challenges such as follows :

- Maintaining accurate records of consumer level electricity usage
- Generating correct and consistent bills from meter readings
- Analyzing electricity patterns across different areas
- Identifying potential power losses due to technical or distribution problems.
- Ensuring data consistency during concurrent meter reading and billing operations

In many existing systems, such data is either fragmented across multiple sources or stored without a proper relational structure, making automation, auditing, and analytics difficult.

This project addresses these challenges by providing a centralized, well-normalized relational database that ensures data correctness, transactional consistency, and efficient querying.

Business Requirements

1. Grid and Area Management

- The system shall store information about power grids and distribution areas.
- Each distribution area shall be connected to exactly one power grid.
- Each power grid may supply electricity to multiple distribution areas.
- The system shall record the data related to total electricity units supplied to each area per month etc.

2. Consumer Management

- The system shall maintain consumer details that include:
 - Consumer ID
 - Name
 - address
 - Associated distribution area
 - Connection type (Residential / Commercial / Industrial)
- Each consumer shall belong to exactly one distribution area and one smart meter.

3. Smart Meter and Meter Readings

- The system shall store smart meter information such as:
 - Meter ID
 - Installation date
 - Operational status
- Monthly meter readings shall be recorded for each meter.
- Integrity constraints shall ensure:
 - Current meter reading $\geq$ previous meter reading
 - Units consumed = Current reading – Previous reading

4. Billing Management

- The system shall maintain tariff slabs based on:
 - Consumer category
 - Unit consumption ranges

- Predefined costs
- Electricity bills shall be automatically generated based on:
 - Units consumed
 - Applicable tariff slab
- Each bill shall store:
 - Billing period
 - Units consumed
 - Bill amount
 - Payment status

5. Consumption and Loss Analysis

The system shall support analytical queries to:

- Calculate total electricity consumption per area
- Compare electricity supplied vs electricity consumed
- Identify high-consumption areas
- Detect areas with potential power losses
- More analytical queries could be added in future.

6. Transactions and Consistency

- Meter reading insertion and bill generation shall occur within a single database transaction.
- If bill generation fails, the meter reading insertion must be rolled back to the last stored state.
- Concurrent billing operations shall preserve database consistency.

Key Design Focus

This project emphasizes:

- Proper relational modeling and normalization
- Strong integrity constraints
- Use of triggers for automated bill generation
- Use of transactions to handle conflicting operations
- Writing analytical SQL queries relevant to real-world utility operations

Conclusion

The Smart Power Distribution & Consumption Analytics System presents a practical and well-structured database solution that closely reflects real-world electricity distribution and consumption management. The project effectively applies core database concepts such as relational modeling, normalization, integrity constraints, transactions, and analytical querying. Its manageable complexity makes it suitable for academic evaluation, while its design also allows future enhancements such as dynamic tariffs, real-time data integration, and advanced consumption analysis. Overall, the system successfully demonstrates how database management systems can be used to build reliable, consistent, and scalable solutions for real-world utility operations.